

LEC07 - Life, Health, and Disability Insurance

Learning Objectives

1. Define life insurance and describe its purpose and principle
 2. Determine your life insurance needs
 3. Distinguish between the types of life insurance policies and analyze various types of life insurance
 4. Select important provisions in life insurance contracts
 5. Create a plan to buy life insurance
 6. Define health insurance and explain its importance in financial planning
 7. Recognize the need for disability income insurance
 8. Understand the value of supplemental health insurance
-

What is Life Insurance

- Life insurance is obtained by purchasing a policy, with the insurance company promising to pay a lump sum to the person specified (beneficiary) at the time of the insured's death, or sometimes while they are still alive
 - In some policies money is paid to the policy holder (the insured) if they are still alive at a future date
 - Company makes promise to pay in exchange for the payment of a premium
-

Purpose of Life Insurance

- The purpose of life insurance is to protect from financial loss related to your death
 - Proceeds may be used to...
 1. Pay off a home mortgage or other debts at the time of death
 2. Provide lump sum payments through an endowment to children when they reach a specific age
 3. Provide an education of income for your children
 4. Make charitable bequests upon your death
 5. Provide a retirement income
 6. Accumulate savings
 7. Establish a regular income stream for survivors
 8. Set up an estate plan
 9. Make a probate fee or final tax return payment
 - Life insurance is one of the few ways to provide liquidity at the time of death
-

The Principle of Life Insurance

- Mortality tables provide your odds of dying
- Your premium is based on your life expectancy and the projections for the payouts for persons who die

LEC07 - Life, Health, and Disability Insurance

- Adjustable for factors that increase/decrease an individual's risk and various administration fees for the company
-

Example

A group of 100,000 males, age 30, wishes to contribute each year to an amount to a common fund sufficient to pay 1000 to the dependents of each group member who dies during the year. Use the mortality table to determine the following:

- a) How many members of the group can be expected to die during the year
- b) What amount must each of the 100,000 members contribute at the beginning of the year to provide 1,000 for the dependants who die before the end of the year

a) $100000 \cdot 0.00112 = 112 \text{ deaths}$

b) $(112 \cdot 1000) \div 100000 = \1.12

Do You Need Life Insurance

- Do you have people you need to protect financially
 - Do you have a partner who works
-

Determining Your Life Insurance Objectives

1. How much money do you want to leave behind
 2. What income do you need for retirement, have special expenses or want to be debt free
 3. How much will you be able to pay for insurance
-

Estimating Your Life Insurance Requirements

Income Replacement Method

- Typically, you will need 70% of your salary for seven years while family adjusts
- $= \text{current income} \cdot 7 \cdot 0.7$
- Assume a typical family with NO liquid assets available that would otherwise reduce your life insurance needs

March 3, 2025

LEC07 - Life, Health, and Disability Insurance

Nonworking Spouse Method

- Insurance experts have estimated that extra costs of up to \$13,500 a year may be required
 - Minimum requirement
- Multiply the number of years before the youngest child reaches age 18 by \$13,500

Family Need Method

- More thorough than the first method because it also considers employed provided insurance, social security benefits, income, and assets

What you have now (A)	
Bank accounts	
Taxable investments and savings, other than your home and other real estate	
Real estate you own other than your primary residence	
Individual life insurance policies	
Group life insurance policies	
Total of what you have now	
What your dependants may need (B)	
Debt and expenses due at your death	
Funeral and burial expenses	
Final income taxes	
Estate probate and administration costs	
Mortgages and loans not covered by insurance	
Credit card debt	
Subtotal	

LEC07 - Life, Health, and Disability Insurance

Living Expenses	
75% of your total family income (before tax) x the number of years your dependants will need it	
Emergency fund (3-6 months of income)	
Children's education	
Average annual potential cost x average number of years in college x number of children	
Total of what your dependants may need	
Total of what your dependants may need (B)	
Total of what you have now (A)	
How much estimated life insurance you need (B - A)	
If C is positive, you may need more life insurance	
If C is negative and your estimates are realistic, your dependants may be able to manage without additional life insurance	

Example

You are the sole wage earner in a typical family, with \$80,000 gross annual income. Use the income replacement method to determine how much life insurance you should carry?

$$= 80000 \cdot 7 \cdot 0.7 = \$392,000$$

Example

You have a child age 5 and are a stay-at-home parent. Using the nonworking spouse method, how much coverage would approximately be needed?

$$= 13500 \cdot (18 - 5) = 13500 \cdot 13 = \$175,500$$

LEC07 - Life, Health, and Disability Insurance**Example**

You and your spouse are in good health and have reasonably secure careers. You make about \$85,000 annually and have opted for life insurance coverage of two times your salary through your employer. With your spouse's income, you are able to absorb ongoing living costs of \$45,000 a year. You own a home with a \$480,000 mortgage. Other debts include a \$15,000 car loan, \$5,000 student loan, and \$3,000 charged to credit cards. In the event of your death, you wish to leave your family debt-free. One of your most important financial goals involves building an education fund of \$80,000 to cover the cost of a four-year university program for your two children ages 2 and 4. To date, you have accumulated \$15,000 toward this goal in an RESP. Should you die, your beneficiaries would receive a \$2,500 death benefit lump sum payment from the Canada Pension Plan. You also have \$25,000 in your company defined contribution pension plan. Average funeral expense is \$15,000. Your other financial assets are as follows:

- Bank accounts - \$4,100
- Term deposits (3 months) - \$3,000
- TFSA High Interest Savings - \$5,000
- Stock investment account - \$2,500
- RRSP's - \$10,500

Use the family need method to determine your life insurance needs.

What you have now:

Bank Account	+ \$4,100
Term Deposit	+ \$3,000
TFSA	+ \$5,000
CPP Death Benefit	+ \$2,500
Stock Investment	+ \$2,500
Company (Group) Life Insurance (2x gross income)	+ \$170,000
TOTAL	= \$187,100

What your dependants may need

Assume 75% · \$85,000 · 7 years	+ \$446,250
Education fund (\$80,000 – \$15,000)	+ \$65,000
Debt repayments	
Mortgage	+ \$480,000
Car Loan	+ \$15,000
Student Loan	+ \$5,000
Credit Cards	+ \$3,000
Emergency fund (assume 3 months of income)	+ \$21,250
Funeral Expenses	+ \$15,000
TOTAL	= \$1,050,500

LEC07 - Life, Health, and Disability Insurance

Amount needed

TOTAL (Represents additional insurance needs) = \$863,400

Example

Analyze the three methods of determining life insurance requirements. Which method is best, and why?

Income Replacement Method

- Simple and easy
- May not provide enough income

Nonworking Spouse Method

- Is a basis to start from as in reality the actual salary of what a stay at home parent does actually deserve is probably higher

Family Needs Method

- More accurate and comprehensive
 - Takes more into account
 - Determines insurance based on inflation rate and expected rate of return
-

Factors That Affect Life Insurance Fees

1. The amount of insurance
 2. Your health class
 3. Age
 4. Gender
 5. Marital status
 6. Occupation
 7. Recreation
-

Sources of Disability Income

Disability Insurance: Covers financial loss in the event you become disabled or unable to work

LEC07 - Life, Health, and Disability Insurance

- Employer
 - Paid for by employer
 - Private
 - You pay a premium like other forms of insurance
 - Public (Provincial and Federal)
 - Employment insurance
 - Canada and Quebec Pension Plans
 - Workers' Compensation
 - Short-term or Long-term Welfare
-

Determining Your Disability Income Insurance Requirements

- Determine benefits from all public / private sources
 - Determine short fall based on 70-70% of your after-tax income
 - Consider waiting period and benefit period
-

Example

Kennedi Stapleton, a widow, has a gross pay of \$6,500 a month. Her disability insurance coverage replace 70% of her earnings after a 90 day (3 month) waiting period. However, depending on who is making the premium payments, the replacement earnings may be taxable. What amount would she receive in disability benefits if an illness kept her off work for 270 days (9 months)? What is her disability started when she was age 35 and was severe enough that it lasted until she was 65 years old?

9 Months

$$= 6 \text{ months} \cdot \$6,500 \cdot 0.7 = \$27,300$$

30 Years

$$= (30 \cdot 12 - 3) \text{ months} \cdot \$6,500 \cdot 0.7 = \$1,624,350$$